TREEWALK NYC • STEAM LESSON PLAN

American Linden (Basswood)

Tilia americana

Species background: Native to eastern and central North America, also called American basswood; historically an important tree for Indigenous communities, who used its fibrous inner bark for cordage and its soft wood for carving. Its leaves are the largest of the linden species commonly planted in cities, often four to six inches long, with a lopsided heart-shaped base that's easy to feel with your fingers.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

American Linden (Basswood): Find American linden's big lopsided heart-shaped leaf.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, crayons.

Teacher background

My leaves are some of the biggest lopsided hearts you'll find! In summer, smell my sweet flowers -- bees definitely will.

Procedure

1. Engage: Ask students to find the biggest heart shape they can imagine.
2. Explore: Compare this leaf's size to a student's hand.
3. Explain: Explain that this tree has the biggest heart-shaped leaves of the lindens in the city.
4. Art: Find the biggest heart-shaped leaf and trace it.
5. Evaluate: Student can describe the leaf as a big, lopsided heart.

Discussion questions

What's the biggest leaf you've ever seen?

Assessment

Correctly describes the leaf as heart-shaped and large.

Extension

Compare leaf sizes across different trees on your walk.

GRADE BAND 3-5

American Linden (Basswood): Describe how linden's soft wood has been historically used.

Standards connection: NGSS 3-LS4-2; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, colored pencils.

Teacher background

Smell a linden flower cluster in summer -- bees love the scent. Its soft, light wood has long been favored by woodcarvers because it carves cleanly.

Procedure

1. Engage: Ask what makes some materials easier to carve or shape than others.
2. Explore: If possible, smell the flowers or observe a photo of carved basswood.
3. Explain: Discuss why soft, even-grained wood is prized for detailed carving.
4. Art: Press a real fallen leaf between two pieces of paper and rub the side of a crayon over it to reveal its veins.
5. Evaluate: Student names one traditional use of this tree's wood.

Discussion questions

What everyday carved wooden objects might have come from a tree like this?

Assessment

Correctly names carving or a related wood use.

Extension

Research one object traditionally carved from basswood.

GRADE BAND 6-8

American Linden (Basswood): Explain nectar reward economics and this tree's Indigenous historical uses.

Standards connection: NGSS MS-LS2-5; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, leaf-vein rubbing materials.

Teacher background

Linden nectar is productive enough to be a prized single-source honey -- a good entry point into pollinator-plant mutualism. Indigenous communities historically used the tree's fibrous inner bark for cordage and its soft wood for carving.

Procedure

1. Engage: What does it mean for a plant and a pollinator to have a mutualistic relationship?
2. Explore: Research how basswood's inner bark fiber was used historically for cordage.
3. Explain: Connect nectar economics to pollinator mutualism, and bark fiber use to Indigenous ecological knowledge.
4. Art: Create a leaf-vein rubbing series from 3 leaves, then research and label the vein pattern type (pinnate) and what it tells you about how nutrients move through the leaf.
5. Evaluate: Write a paragraph connecting both the ecological (nectar) and cultural (fiber/wood) uses of this tree.

Discussion questions

What does it mean for a tree to have both an ecological role and a long human-use history at the same time?

Assessment

Paragraph correctly addresses both nectar economics and historical human use.

Extension

Research one other tree species with a documented Indigenous cordage or fiber use.

GRADE BAND 9-12

American Linden (Basswood): Evaluate nectar-volume measurement methods used in pollination ecology research.

Standards connection: NGSS HS-LS2-7; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, research sources, macro photography or sketching tools.

Teacher background

Linden honey is prized enough to be sold as a distinct varietal; research how nectar volume and sugar concentration are measured in pollination-ecology studies, and explain why a single tree species blooming in mass can support a specialized honey market.

Procedure

1. Engage: How would a researcher actually measure the volume and sugar concentration of nectar in a single small flower?
2. Explore: Research real methods used in pollination-ecology field studies (micro-capillary tubes, refractometers).
3. Explain: Connect measured nectar productivity to the existence of single-source honey markets.
4. Art: Photograph or sketch the vein network at close range and compare its branching pattern to a river delta or lung bronchi -- write a short reflection on convergent branching patterns in nature.
5. Evaluate: Submit a written summary of nectar-measurement methodology and the vein-network artwork with reflection.

Discussion questions

What does the branching-pattern similarity between leaf veins, river deltas, and lungs suggest about the physics of efficient distribution networks?

Assessment

Rubric covering methodology research accuracy and depth of the convergent-pattern reflection.

Extension

Compare American linden's nectar economics to littleleaf linden's documented bee die-off research question.

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.